TTG+

Construction savings are clear; DB results depend on the workload and eligible region.

Five existing applications · six fixed fixtures · all 18 cells with n = 3

LinkedList removes per-hop SQL versus Off; both TTG+ versions use16 per Traverse.

New versus850: workflow medians vary; LittleX, IssueDesk and KnowledgeMap remain slower than Off.

Construction microbench: ~1.5–1.8ms AST → ~4.7–7.0µs warm select/bind CPU.

One bound seed/demand origin per template query; four-origin batching is a separate native test.

Research problem: repeated runtime work

Graph traversal interleaves path discovery, payload loading and application state.

Native visit requests the next ordered graph path, then materializes its target payloads.

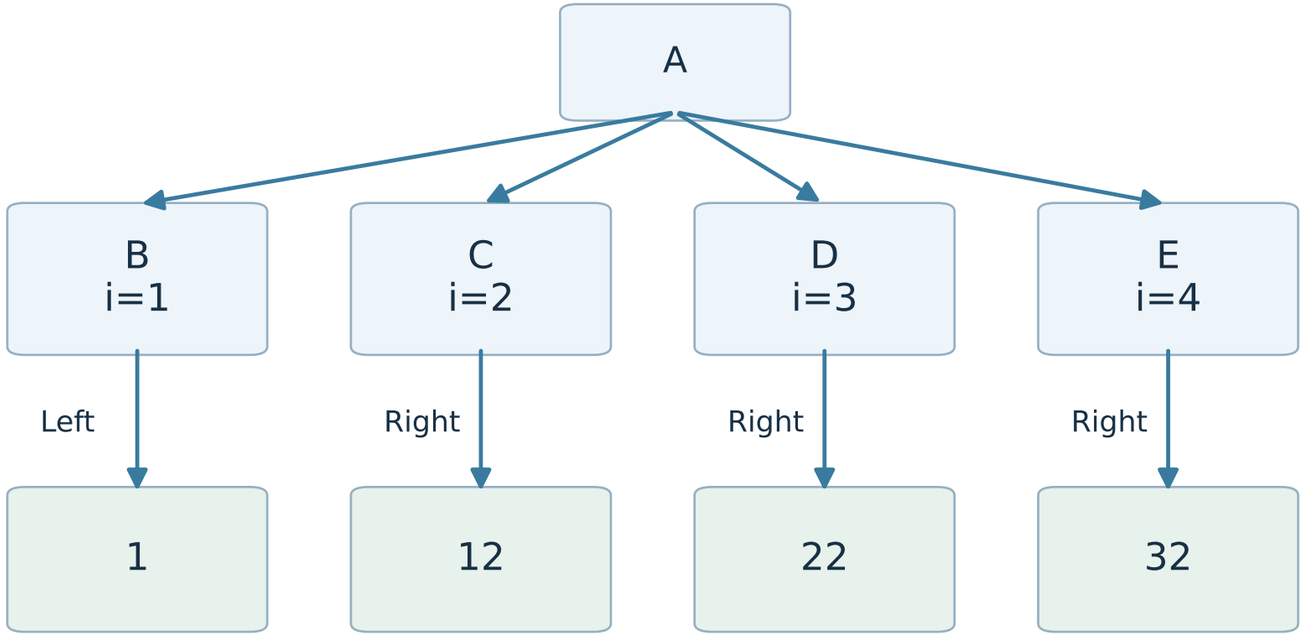
Repeated hops can repeat SQL dispatches; runtime SQL AST construction adds client CPU work.

Future graph accesses depend on program position, guards, carried state and the pending FIFO queue.

TTG+ combines queued continuation analysis, batched path SQL and compiler-emitted typed templates.

At B: continue the known FIFO queue and carry state

A has already queued B, C, D, E. Hook B observes self.i=0 and pending=[C,D,E].



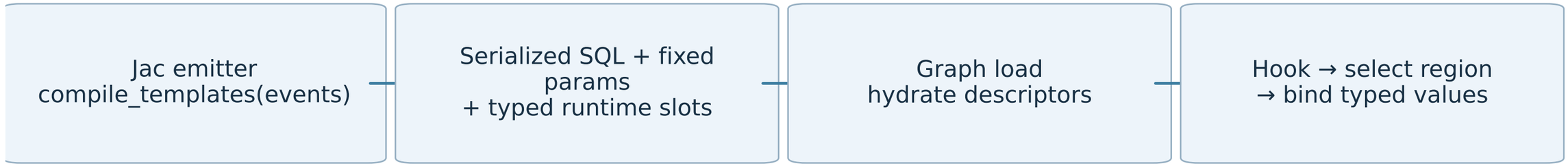

```
can branch with FrontierBranch entry {
self.i += 1;
if self.i == 1 {
visit [->:FrontierLeft:->
[?:FrontierLeaf]];
} else {
visit [->:FrontierRight:->
[?:FrontierLeaf]];
}
}
```

Prediction carries scalar state in FIFO order; native abilities and the native queue retain their execution order.

Graph uses short node names; code uses FrontierWalk native syntax. Reports [1,12,22,32] and final i = 4 match.

Compile once: emit SQL, then hydrate/select/bind

SQL AST generation moves into the actual Jac compiler emitter for covered emitted regions.



Runtime slots: origins per site, actor UUID/text, row limit; linear regions also bind source type/module and depth.

Bound SQL → unchanged PgWire boundary → ordered per-origin path cache + merged process payload loading.

Covered primary paths report template_queries > 0 / legacy_fusion_calls = 0. Ordinary fallback still constructs its own queries.

One emitted template, many runtime origins

Simplified SQL shape; actual SQL and exact fingerprints are retained in the sealed evidence.

```
SELECT slot, origin_order, target_id, arity
FROM unnest(:origins_1::uuid[]) WITH ORDINALITY AS seed
LEFT JOIN LATERAL (native ordered Post → Tweet path) ON TRUE
UNION ALL
SELECT ... FROM unnest(:origins_2::uuid[]) ...
ORDER BY slot, depth, ord
LIMIT :row_limit;
```

Linear form uses a recursive origin_id/depth CTE, continuing only while actual arity == 1 and type/module match.

The analyzer supplies permitted program transitions; SQL checks graph cardinality. Ordered and empty results seed native path cache.

9282 also changes the emitted type-membership predicate to uncorrelated IN. This is a version/package comparison.

Four-origin capability and the disengage assumption

Prediction assumes queued nodes trigger; native execution and cancellation remain authoritative.

Accepted native case	Baseline	Next	Native result
Four-origin FrontierWalk	8 submitted 8 promoted	8 submitted 8 promoted	B,C,D,E; i=4 reports1,12,22,32
FrontierCancel	5 submitted 5 promoted / 0 unused	7 submitted 5 promoted / 2 unused	entered[0,1] reports[] unchanged

The cancelled case demonstrates two extra predicted descendant payloads in this identity-bound fixture.

Queued parent endpoints were already demanded by native visit evaluation. Ability entry and payload demand are distinct.

Why these existing applications

The selection was fixed before timing; unsupported and slower cells are retained.

Application	Native behavior and exercised oracle
LinkedList32 /1000	Traverse + Sum + Find(head/middle/tail/missing); exact order/count/state
LittleX original	Feed / search / likes / follow / create; feed order, authorship, graph and restart
IssueDesk	Engineer then issue visits; assignees/blockers/comments; mutations + cycle rejection
KnowledgeMap	Shared-topic graph, a↔b links and explicit seen guard; publish/review/revise + denials
JDrive	Folder depth/path dictionaries and trashed predicate; tree/trash/mkdir/persistence

Existing accepted Seed 42 fixtures, original native workflows and independent correctness oracles.

Complete fixed selection: DB protocol time

Whole-workflow median and [min, max] ms; all counts are fresh workflows, n = 3 per cell.

Fixture	Off (9282)	TTG+ 85026d	TTG+ 9282
LinkedList · 32	115.081 [113.695, 119.006]	94.461 [94.299, 96.609]	94.371 [93.375, 94.836]
LinkedList · 1,000	3337.873 [3147.031, 3387.414]	2360.957 [2356.131, 2374.843]	2310.506 [2250.325, 2311.392]
LittleX	42.578 [38.132, 43.135]	55.479 [54.380, 56.864]	53.566 [51.473, 54.371]
IssueDesk	139.440 [138.211, 142.840]	154.267 [152.613, 157.884]	159.419 [157.615, 163.092]
KnowledgeMap	72.712 [71.422, 73.791]	78.997 [76.774, 84.614]	79.167 [79.033, 79.360]
JDrive	48.053 [45.434, 48.143]	46.784 [46.523, 47.214]	46.648 [45.128, 46.885]

All 72 observed protocol sums equal interval unions. Timing includes actual data and transaction-control SQL.

All SQL is counted: data + control = TOTAL

Whole workflow; every count is identical across its three measured repeats.

Fixture	Off	85026d	9282
LinkedList · 32	318 + 42 = 360	196 + 47 = 243	196 + 47 = 243
LinkedList · 1,000	9030 + 42 = 9072	5036 + 47 = 5083	5036 + 47 = 5083
LittleX	90 + 63 = 153	78 + 69 = 147	82 + 69 = 151
IssueDesk	368 + 189 = 557	368 + 193 = 561	368 + 193 = 561
KnowledgeMap	186 + 112 = 298	186 + 115 = 301	186 + 115 = 301
JDrive	107 + 49 = 156	107 + 49 = 156	107 + 49 = 156

Foreground, child payload reads, path discovery, identity/catalog and transaction completion are retained.

Native Traverse: SQL reduction is already present in850

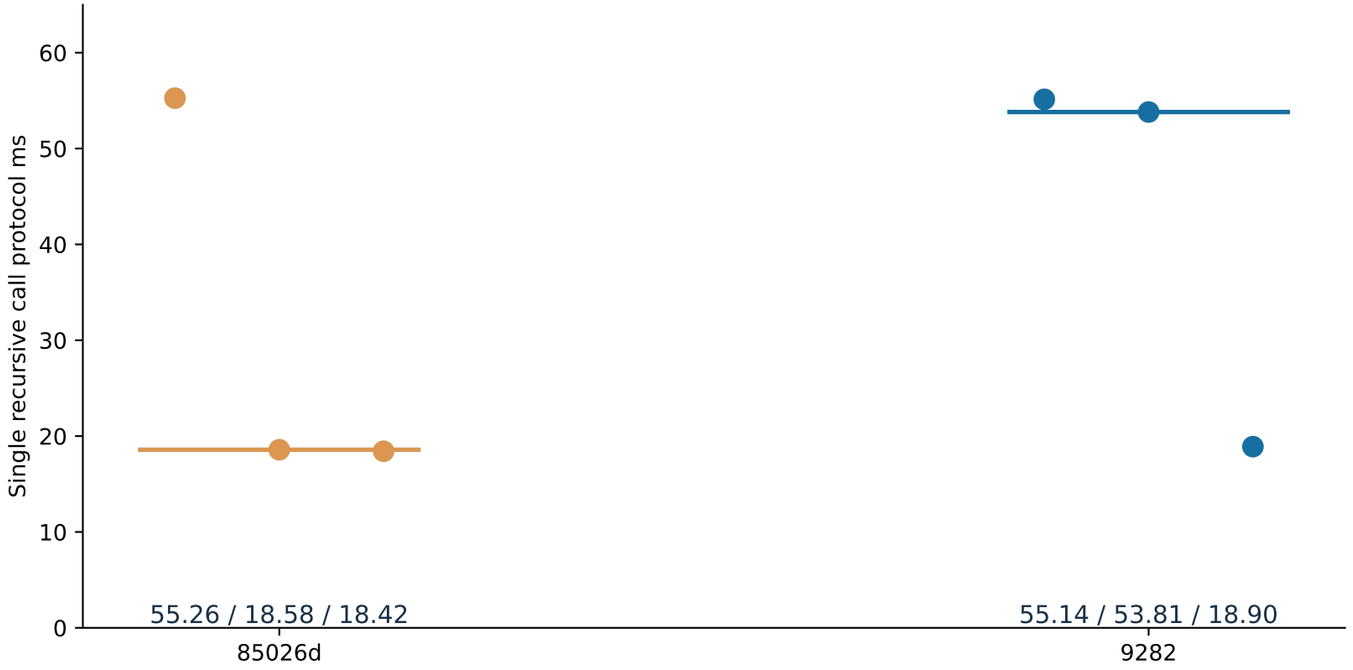
TOTAL includes path discovery, payload reads, identity/catalog and controls; all n = 3 counts constant.

Fixture /mode	Protocol ms median [range]	Path	Payload	Control	TOTAL
LinkedList · 32 /Off (9282)	25.357 [23.573, 26.281]	32	35	7	76
LinkedList · 32 /TTG+ 85026d	7.056 [6.985, 7.329]	1	5	8	16
LinkedList · 32 /TTG+ 9282	7.246 [6.585, 7.841]	1	5	8	16
LinkedList · 1,000 /Off (9282)	733.271 [719.127, 751.061]	1000	1003	7	2012
LinkedList · 1,000 /TTG+ 85026d	31.639 [31.164, 68.167]	1	5	8	16
LinkedList · 1,000 /TTG+ 9282	66.599 [31.942, 68.083]	1	5	8	16

Every row also has1 identity and 1 catalog query. Fused path counts 1; Off ordinary paths32/1000. Exact visited32/1000.

LL1000: the recursive SQL interval carries the variation

Same 1,000 returned rows /161,806 DataRow payload bytes; one recursive call per Traverse.



850:55.257 /18.580 /18.421 ms
9282:55.135 /53.809 /18.901 ms

Payload 5 calls stay around 10.6–11.1 ms.
Both versions have slow and fast samples.
Primary per-run plans/statistics are unavailable.

Workflow medians: 3,337.873 ms Off → 2,360.957 ms (850) → 2,310.506 ms (9282); Traverse medians differ in the opposite direction.

Matched recursive SQL: accepted native recaptures

Two full six-operation workflows; exact captured SQL/Bind/state and original 1,000-row completion.

Version	Planning ms: first / repeat	Execution ms: first / repeat	Root shared hits
TTG+ 85026d	1.190 / 0.669	52.716 / 49.895	20136
TTG+ 9282	0.695 / 0.578	63.194 / 62.847	20136

Serializable transactions; statistics rows = 0 and reltuples = −1. Census and EXPLAIN warm buffers.

Shared reads = 0; spills = 0; JIT absent. Root buffer counters are direct inclusive values.

Diagnostic EXPLAIN: first/repeat measurements, separate from the primary n = 3 protocol samples.

Controlled ANALYZE changes the plan and execution cost

Restored captured relevant state, original DDL/indexes/settings and exact Bind; same ordered 1,000-row result.

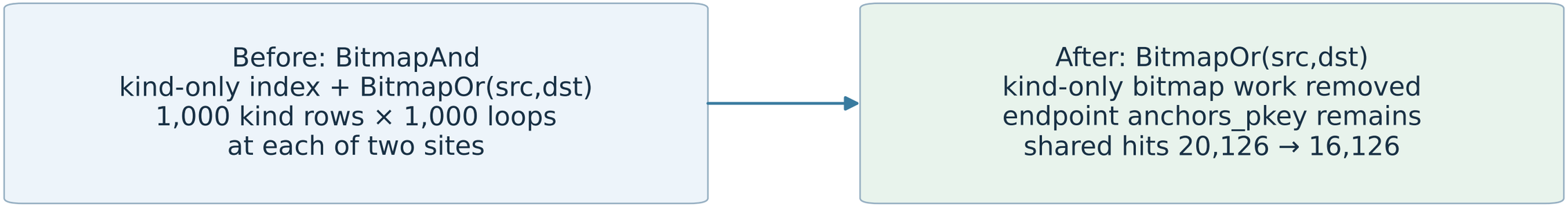
Version	Statistics regime	Planning ms first/repeat	Execution ms first/repeat	Shared hits
TTG+ 85026d	Before ANALYZE	0.611 / 0.601	47.970 / 48.328	20126
TTG+ 85026d	After ANALYZE	0.552 / 0.572	13.386 / 13.413	16126
TTG+ 9282	Before ANALYZE	0.632 / 0.593	49.923 / 49.434	20126
TTG+ 9282	After ANALYZE	0.533 / 0.530	13.721 / 13.382	16126

Before: reltuples = −1 for anchors/types. After: anchors = 2,005, graph_types = 4. Shared reads = 0.

Controlled restored fixtures establish statistics sensitivity. Primary per-run plans/statistics are unavailable.

Why ANALYZE helps in the controlled fixture

Measured operator change at two path-evaluation sites in the controlled restored fixture.



Execution falls from about 48–50 ms to 13.4–13.7 ms on both restored versions; JIT is absent in all 12 diagnostic plans.

Controlled restored fixture: captured relevant columns, original defaults for unused columns, fresh heap, warm buffers.

Next work: explicit statistics regimes, cost admission and database prepared-statement reuse. Primary per-run statistics unavailable.

LittleX: four feed requests add four data calls in9282

Each feed:850 has 2 fused queries;9282 has 1 fused root query +2 ordinary collect paths.

Whole workflow	Off	85026d	9282
Protocol ms median	42.578	55.479	53.566
Data +control =TOTAL	90 +63 =153	78 +69 =147	82 +69 =151
Per feed path discovery	3 ordinary	1 root +1 batch	1 root +2 ordinary
Per feed payload reads	6	4	4

The four extra calls are path discovery; payload reads remain four per feed. Native outputs, order and final graph match.

LittleX limiter: one frontier attempt per continuation invocation

Source-backed interpretation, joined to compact counters, emitted event IDs and actual SQL sequence.

```
frontier_attempted = []
resolve(active, event):
if cached_path: return cached_path
if graph.sql_templates and not frontier_attempted:
frontier_attempted.append(True)
resolve_frontier(...)
if path_missing: resolve_ttg_path(...)
```

Event 1: Root → Profile
consumes the one attempt

Events 4/5: collect visits
ordinary fallback and reuse cache

Later hooks return reused paths
before another frontier attempt

Recorded: template query 1, linear region 1, fallback resolver 2 and reached collect events 4/5; source-backed control interpretation.

Gather 277 is a real tail barrier after both visits. It is insufficient to explain the missing current-ability batch.

Proposed, untested: retry distinct proven regions/activations with path reuse/dedup; preserve guard and native semantics.

One bound seed/demand origin per template query

Fixed application queries use one bound seed/demand origin; a separate native integration binds four.

Fixture	Template SQL	Bound seed/demand origins	Submitted	Promoted
LinkedList · 32	81	81	140	140
LinkedList · 1,000	2501	2501	4496	4496
LittleX	6	6	63	63
IssueDesk	9	9	14	14
KnowledgeMap	4	4	9	9
JDrive	0	0	0	0

One seed recursively covers 1,000 LinkedList steps. Cross-queued-origin demand width remains one in the fixed applications.

Workflow counts are constant across n = 3; submission/promotion totals sum native per-scope ID sets.

Domain cases: reached barriers and controls remain visible

Actual compact checkpoint counts bind reached source locations and outcomes.

Application	Observed evidence	Measured result
IssueDesk	9 template queries;14 submitted/promoted project:98, engineer:104, issue:109 barriers reached	561SQL vs557 Off 159.419ms vs139.440ms
KnowledgeMap	4 templates;9 submitted/promoted article:52 / article:171 and start:165 barriers	301SQL vs298 Off 79.167ms vs72.712 ms
JDrive	0 template queries /0 submissions walk:45 /walk:48 /run:41 barriers reached	156SQL in all modes 46.648ms vs48.053ms Off

Mutations, nested spawns, cycles/private views and restart/persistence were retained in independent stored replay and physical state checks.

Construction-only thread CPU: AST versus templates

Construction microbenchmark: 100 calls/block × 3 blocks; 10 excluded AST/warm priming calls.

Module/shape	850 AST µs/call	9282 cold hydrate µs	9282 warm select+bind µs
LinkedList / linear	1468.939 [1463.568, 1546.886]	52.568 [51.907, 53.801]	4.705 [4.704, 4.737]
LittleX / collect batch	1843.307 [1841.916, 1848.335]	125.383 [124.803, 127.044]	7.033 [7.026, 7.058]

AST build count: 100/block for dynamic construction; 0 for hydration/selection/binding. Same-thread clock controls passed.

Cold hydration covers the selected walker bundle. Inputs bind actual emitted sites and fixture IDs with declared common2048 limits.

Declared LittleX collect-batch microbenchmark; the primary 9282 feed uses root linear.

Selection and binding are measured separately

Runtime only: 100 calls/block × 3 blocks; compiler measurements use 1 compile/block.

Declared emitted shape	Select µs/call	Bind µs/call	Select +bind µs/call
LinkedList / linear	0.309 [0.305, 0.329]	4.449 [4.408, 4.449]	4.705 [4.704, 4.737]
LittleX / collect batch	0.217 [0.216, 0.221]	6.861 [6.776, 7.025]	7.033 [7.026, 7.058]

Separate select/bind counters use opt-in timers; unprofiled blocks retain native clocks disabled.

Cold selected-walker hydration is separate; copying/spec validation/artifact metadata stay outside measured blocks.

LL uses a covered linear shape. LittleX uses declared collect events 4/5; primary9282 selects only root linear.

Compilation and artifact growth are part of the tradeoff

First compiler calls include lazy initialization; full CPU vectors and controls are preserved.

Module	Same-program CPU ms 850 /9282	Generated Python bytes	Bytecode bytes
LinkedList module	250.7 [250.5, 267.4] 242.2 [241.9, 245.8]	45682 → 219988	36136 → 90124
LittleX module	548.6 [503.0, 1064.3] 537.7 [536.7, 1051.7]	217118 → 1499200	122908 → 280609

Same-program runs follow one excluded compile; n = 3. First fresh-program CPU: LL ≈ 3.9/3.8s; LittleX ≈ 4.6/4.5s (850/9282).

All fresh/same-program, wall/thread CPU, emitted SQL bytes and opt-in generation counters remain in the editable data.

Full compiler vectors retain initialization and dispersion; generated Python and bytecode artifacts are larger.

SQL generation: measured cost and size

Actual emitter generation timers; separate profiled and unprofiled controls.

Module	Generation calls	Generation thread CPU ms	SQL bytes	Bundle bytes
LinkedList module	6	11.901 [11.874, 12.477]	42360	53709
LittleX module	24	33.893 [33.357, 34.078]	124565	156172

Counts include all walker checkpoint emission in the compiled module; bundles include catalog, batch, linear and native descriptors.

SQL generation occurs during compilation; emitted descriptors, generated Python and bytecode sizes are measured.

Real process readers; zero observed SQL interval overlap

A child process existing is insufficient: actual PgWire calls and distinct backend PIDs are recorded.

LL1000 representative Traverse: foreground PID [2931381]; child PID [2931673].

14 foreground SQL +2 child SQL; actual separate PostgreSQL backends.

All 72 protocol sums equal interval unions; observed SQL overlap = 0.

Python 3.14 has the GIL enabled; actual child SQL executes on independent PostgreSQL backends.

Startup/readiness and IPC remain in the natural workflow; DB totals cover only their actual SQL calls.

Measurement and lifecycle

Primary clock: driver/socket sendall to ReadyForQuery, including response/row-decoding progress.

Clock immediately before real sendall → receipt of ReadyForQuery Z; row decoding/progress included.

Parameter encoding/compiler/model work stays outside that interval; outer driver-call times retained separately.

C CommandComplete, D rows/bytes, actual command cycles and SQL fingerprints retained; controls count.

Fresh per-trial DB; identical seeded actions; one profile reset after setup; Always/process 1; limit 2048/steps 32768.

OS/PG buffers warmed by seed/verification; SQL statement logging off; WAL min 32 / max 64 MB.

What to improve next

Proposals are untested improvements, selected from measured limits rather than assumed speedups.

Region-aware retry/dedup: remove the single-attempt LittleX limitation while preserving native cache reuse.

Admission by opportunity and cost: wider eligible origins matter; current application queries observe width1.

Deferred roadmap: cost-model admission and database prepared-statement reuse, evaluated against workload selectivity.

Artifact budget: consider deduplicated descriptors, lazy region hydration and selective template emission.

Retain guard/identity/oracle checks and intentional overfetch accounting; evaluate existing workflows without dropping regressions.

Reproducible evidence and exact completed inventory

72/72 primary workflows; 18/18 native correctness cases; five existing applications.

Runtime versions85026d and9282e246; primary Off uses9282. Same process 1/2048/32768 settings.

54 measured +18 excluded warmups; all 18 n = 3 cells, complete CSV/JSON, sealed SQL/HTTP/events and semantic proofs.

Independent audit:142,488 driver records /3,184 sealed files; all request/count/interval/export statistics match.

Actual compiler output/artifacts and separate CPU controls; native four-origin/cancel evidence; source-bound LittleX explanation.

Editable renderer + figure inputs + report/source hashes. Primary and diagnostic evidence remain separately identified.